Against Distributed Deletion

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Abstract

This paper examines a previously unnoticed 'split' construction in Greek, where a possessor that originates in a PP occurs together with the P separated from the possessum. I show a correlation between the availability of this split and the interpretation of a PP. This finding poses a challenge to PF-based accounts of splitting, particularly those that assume distributed deletion (e.g., Fanselow and Ćavar 2002; Fanselow and Féry 2006, Bondarenko and Davis 2023, Murphy and Wilson 2025 i.a.). Such accounts require additional mechanisms beyond those independently required by a syntactic account and fail to predict the distribution of split constructions. Instead, I propose a purely syntactic analysis that accounts for splits in Greek based on a correlation between the interpretation of a PP and its merge height (e.g., Alexiadou and Anagnostopoulou 2007, Cinque 1999, Schweikert 2005). These findings together with additional theoretical considerations will be shown to provide strong support for the elimination of distributed deletion as a mechanism in natural language.

1 Introduction

This paper investigates a previously unexplored split in Greek that involves possessor constructions embedded under Ps. In Greek possessor constructions, a genitive possessor, *pjanu* 'whose,' can be separated from a possessum, *to podhilato* 'the bike' (Horrocks and Stavrou 1987, Mathieu and Sitaridou 2005, Alexiadou et al. 2007, Angelopoulos 2019, Angelopoulos and Michelioudakis 2024 i.a.):

(1) Pjanu$_i$ idhe to podhilato t$_i$?
whose.GEN saw.3SG the bike.ACC
'Whose bike did she see?'

I present an asymmetry split within PPs. Although a possessor can never be separated from a possessum embedded under a P, (2a), some PPs allow what appears to be the movement of a non-constituent in (2b). In this pattern (hereafter, PP-split), the possessor and the P co-occur in the left periphery to the exclusion of the possessum, which is left stranded postverbally. An important finding of this paper is that PP-splitting is not allowed with all PPs. Instead, its availability depends on the specific interpretation of the PP. For instance, while PPs expressing subject matter allow for splitting, as we saw in (2b), PPs with an evidential interpretation block both standard possessor movement and the formation of a PP-split, (3).

(2) a. * Pjanu$_i$ harike ja tin epitihia t$_i$?
whose.GEN was.happy.3SG for the success.ACC
'For whose success was she happy?'

b. [Ja t$_i$ pjanu]$_j$ harike tin epithia$_i$ t$_j$?
for whose.GEN was.happy.3SG the success.ACC

intended: 'For whose success was she happy?

(3) a. * Pjanu$_j$ efije simfona me/ kata tin adherfi$_i$ t$_j$?
whose.GEN left.3SG according to according to the sister.ACC
'According to whose sister did she leave?'

b. * [Simfona me/ kata t$_i$ pjanu]$_j$ efije tin adherfi$_i$ t$_j$?
according to according to whose.GEN left.3SG the sister.ACC

intended: 'According to whose sister did she leave?'

Similar splits are found across a different languages, particularly within Slavic languages.[1] These splits have given rise to different analyses, with important implications for our understanding of syntactic structure and the division of labor between syntax and phonology. Syntactic accounts of these splits have been proposed in Franks and Progovac (1994); Kayne (2002); Bašić (2008, 2009); Abels (2003, 2012); Talić (2019) and Goncharov (2015). In contrast, proponents of approaches employing mechanisms like distributed deletion that selectively delete copies at PF, include Fanselow and Ćavar (2002); Bošković (2015); Fanselow and Féry (2006); Bondarenko and Davis (2023), among others. Crucially, the syntactic approach finds support in the observation that the interpretation of the PP directly affects its potential for splitting, as it aligns with established findings that

[1] Besides Modern Greek (see also Androutsopoulou 1997), left-branch extractions appear in Ancient Greek (Mathieu and Sitaridou 2005), Hungarian (Szabolcsi 1983) and Ch'ol (Little 2020), while French displays *combien*-splits (Starke 2001; Kayne 2002). Germanic languages also exhibit comparable phenomena, such as *was für*-splits in German (Abels 2003; Leu 2008) and *wat voor*-splits in Dutch (den Besten 1985; Corver 2017). Additionally, discontinuous noun phrases are documented in languages like Bosnian/Croatian/Serbian (Talić 2019), Russian (Pereltsvaig 2008, Bondarenko and Davis 2023), Mohawk (Baker 1996) and Iquito (Murphy and Wilson 2025).

PPs occupy different syntactic positions depending on their interpretation (Alexiadou and Anagnostopoulou 2007, Cinque 2006, Schweikert 2005 i.a.). Distributed deletion, however, faces difficulties with PP-splits, requiring principles beyond those of a purely syntactic analysis to account for their distribution. Moreover, when applied to Greek, it predicts ungrammatical structures. Consequently, a syntactic analysis of Greek PP-splits provides evidence against distributed deletion as a mechanism. To formalize this, this paper argues that there are principles which effectively prohibit distributed deletion in grammar, as summarized in (4). This has far-reaching consequences, calling for a re-examination of other splits in other languages (see Goncharov 2015). This paper focuses on Greek PP-splits shown in (2b) to demonstrate the issues of distributed deletion and motivate (4).

(4) * **Distributed deletion**: Distributed deletion is never allowed in natural language.

This paper proceeds as follows. Section 2 demonstrates that PP splits in Greek are allowed only with certain Ps, depending on their interpretation. This section reveals the striking behavior of a particular P: it allows or blocks PP-splits based on its interpretation. Building on these empirical observations, Section 3 proposes an analysis of the cases in which a possessor is separated from the possessum in Greek, extending this analysis to account for both the availability and unavailability of PP-splits. Section 4 discusses the numerous challenges faced by distributed deletion accounts to explain this complex pattern. By demonstrating the inadequacy of this approach, this section strengthens the argument for the constraint proposed in (4). It also discusses the principles behind the prohibition of distributed deletion as a mechanism of natural languages. Section 5 concludes.

2 Possessor movement and PP-splits in Greek

I present a study of possessor movement and PP-splits in Greek. I examine a variety of PPs expressing evidential, temporal, benefactive, source, agent, causer, manner, instrument, locative (either denoting a static location or direction), comitative and subject matter (in the sense of Pesetsky 1996). These PPs range in structural complexity, from simple Ps (*kata* 'according to') to complex Ps (*simfona me* 'according to') allowing us to investigate whether structural complexity plays any role in the formation of PP-splits. As we shall see, it does not.

	Evid.	Temp.	Benef.	Sour.	Ag.	Caus.	Man.	Instr.	Loc.	Comit.	SM.
Possessor movement	×	×	×	×	×	×	×	×	×	×	×
PP-splitting	×	×	✓	✓	✓	✓	✓	✓	✓	✓	✓

Table 1: Possessor movement and PP-splitting across PPs

All of the Ps illustrated in the table above can be combined with a possessum that takes accusative case and a dependent possessor that takes genitive case. As shown in (2a), a possessor originating within a PP cannot occur alone in the left periphery of the clause, stranding P and the possessum. This restriction applies to all Ps listed in Table 1, as illustrated in the second row.[2] To further illustrate,

2 The data presented in this paper are based on the author's own grammaticality judgments, which were systematically verified through a questionnaire administered to ten native speakers of Modern Greek (ages 20-65, all from the Athens region). The questionnaire elicited acceptability ratings for various PP-splits, and any instances of speaker variation (observed in very few cases) are explicitly discussed.

similar to the restriction observed in (2a), possessors originating in a benefactive, (5), also cannot appear alone clause initially.

(5) a. Majirepse ja ton adherfo tis Marias.
cooked.3SG for the brother.ACC the Maria.GEN
'She cooked for Maria's brother.'

b. * Pjanu$_i$ majirepse ja ton adherfo t_i?
whose.GEN cooked.3SG for the brother.ACC
intended: 'For whose brother did she cook?'

Conversely, PP-splitting (2b) is permitted with certain types of PPs but not with others. Splitting is allowed with agent, causer, benefactive, instrument, source, locative, comitative, subject matter, and manner PPs (see 7 for an example with agent PPs). However, evidential and temporal PPs do not permit splitting (see 6 and 8, repeated from 3, respectively for examples with temporal and evidential PPs).[3] Beyond the simple Ps discussed so far, such as *ja* in (2b), there are Ps like *simfona me*, which differ from simple Ps in that they are morphologically complex. Like plain Ps, complex Ps can express different interpretations. Interestingly, the availability of PP-splitting for such a P depends on its interpretation. While *simfona me* blocks splitting when used with an evidential meaning, (8), it allows splitting when expressing manner, (9). This contrast shows that a P's capacity to license splitting is determined by its interpretation rather than its structural complexity.[4]

[3] Temporal PPs, as in (6b), are the only type of PP where speaker variation appeared to be a factor. Specifically, two of the ten speakers consulted found PP-splits involving temporal PPs acceptable.

[4] The availability of PP-splits in Greek, where the accusative possessum is separated from the rest of the PP, suggests that Greek PPs are not islands. This is because to derive such splits,

(6) a. Efije prin tin aderfi tu Adrea.
left.3SG before the sister.ACC the Adrea.GEN
'She left before Adrea's sister.'

b. * Prin pjanu efije tin adherfi?
before whose.GEN left.3SG the sister.ACC
'Before whose sister did she leave?'

(7) a. Enohlithike apo ton adherfo tis Eleanas.
got.annoyed.3SG by the brother.ACC the Eleana.GEN
'She got annoyed by Eleana's brother.'

b. [Apo t_i pjanu$_i$]$_j$ enohlithike ton adherfo$_i$ t_j?
by whose.GEN got annoyed.3SG the brother.ACC
intended: 'By whose brother did she get annoyed?'

(8) a. Efije simfona me/ kata tin adherfi$_i$ tis Marias?
left.3SG according to according to the sister.ACC the Maria.GEN
'She has left according to Maria's sister?'

b. * [Simfona me/ kata t_i pjanu]$_j$ efije tin adherfi$_i$
according to according to whose.GEN left.3SG the sister.ACC
t_j?

intended: 'According to whose sister did she leave?'

(9) a. Zi simfona me tis simvules tu patera tu.
live.3SG along with the advice.ACC the father.GEN his.GEN
'He lives along with his father's advice.'

b. [Simfona me t_i pjanu]$_j$ zi tis simvules$_i$ t_j?
along with whose.GEN live.3SG the advice.ACC
intended: 'Along with whose advice does he live?'

Consequently, we have detected two distinct groups of PPs, shown in the pairs in (8) and (9). The group of PPs in (8) is different because it prevents both

movement must take place from within the PP, as will be discussed in the following section.

possessor movement and PP-splitting, as opposed to the group in (9), which only prevents the first. The different behavior of these PPs, and in particular, this of *simfona me* in (8) and (9) suggests that PP-splitting depends on the interpretation of the PP. Having observed the varying behavior of PPs with different interpretations with respect to PP-splitting, I will now demonstrate how this behavior can be captured under a syntactic analysis.

3 A syntactic analysis of splits

I begin with the analysis of plain cases of possessor extraction and then show how the account extends to PP-splits.

3.1 A syntactic analysis of possessor extraction

To understand why PPs behave differently in PP splitting, we first need to analyze how possessors are separated from a simple DP possessum argument, as shown in (10a), repeated from (1).[5] This analysis is defended in length in Angelopoulos (2019) and Angelopoulos and Michelioudakis (2024). Here, I discuss key points of their analysis, and extend it to PPs. To start with, this analysis is based on the observation that possessors in Greek can appear before the noun, as in (10b). In this position, the possessor and the accusative possessum form a constituent,

[5] An alternative analysis based on successive cyclic movement has been proposed by Horrocks and Stavrou (1987). However, this approach faces challenges, as discussed in (11) (see also Angelopoulos 2019 and Angelopoulos and Michelioudakis 2024 for additional challenges).

evidenced by their ability to undergo both short and long-distance movement to the left periphery of the clause, as shown in (10c) with short movement.

(10) a. Pjanu$_i$ aghorase to podhilato t$_i$?
whose.GEN bought.3SG the bike.ACC
'Whose bike did she buy?'

b. Aghorase [pjanu$_i$ to podhilato t$_i$]?
bought.3SG whose.GEN the bike.ACC
intended: 'Whose bike did she buy?'

c. [Pjanu$_i$ to podhilato t$_i$] aghorase?
whose.GEN the bike.ACC bought.3SG
intended: 'Whose bike did she buy?'

This prenominal position, which I label FrontP, is projected in the left periphery of the DP. Similar positions have been identified in various languages, including Gungbe (Aboh 2004), Italian (Cardinaletti 1998; Giusti 2006), Polish (Siewierska and Uhlírová 1998), Turkish (Rijkhoff 1997), Bangla (Syed 2015), among many others. The exact function of FrontP, whether it should be analyzed as FocusP (Ntelitheos 2004) or something else (Szendrői 2010), is not crucial for the current analysis; what matters is that it exists.[6] The existence of this position accounts for the fact that the accusative possessum can be moved on its own, as in (11a). In this case, the possessor moves to Spec, FrontP, as shown in (11b), which allows the possessum to form part of a DP constituent that can subsequently move. In contrast, if prenominal possessors were located in Spec,DP (Horrocks and Stavrou

[6] Several of the examples in this paper feature both alienable and inalienable possessors. This distinction is crucial, as Alexiadou (2008) has shown that they are merged in distinct positions: Inalienable possessors are merged as complements, and alienable possessors as specifiers. Although noun movement in Greek obscures this difference, both types are nevertheless able to move to Spec,FrontP which should then count as a criterial position.

1987), as in (11c), the possessum should not be able to move on its own. This is because intermediate projections like D′, shown in (11c), are generally not movable.[7,8]

(11) a. Pja tenia$_i$ idhes [t$_i$ tu Fellini]?
which movie.ACC saw.2SG the Fellini.GEN
intended: 'Which movie did you of Fellini's?'

b. [$_{FrontP}$ [$_{DP}$ tu Fellini]$_j$ [$_{Front}$ Front′ [$_{DP}$ pja$_D$ tenia t$_j$]]

c. [$_{DP}$ [$_{DP}$ tu Fellini]$_j$ [$_{D'}$ pja$_D$ tenia t$_j$]]

In cases where a possessor is separated from the possessum, as in (10a), I assume that the possessor can never be moved out of the DP to Spec,CP. This restriction stems from the DP acting as a bounding node that blocks extraction out of it to the left periphery (Angelopoulos and Michelioudakis 2024). To be more precise, following Aissen and Polian (2025, (82)) and Angelopoulos and Michelioudakis (under review), I encode the opacity of the DP-domain as a case of "selective opacity" (Keine 2019). In particular, I assume that any phrase in the extended projection of N^0, which also includes D and Front, i.e., Front>D>N, is opaque to a *wh*-/focus/topic/ or relative probe on a c-commanding C^0. In Keine's terms, any node in the extended projection of N^0 is a "horizon" for a *wh*, relative, topic or focus probe on C^0, rendering elements dominated by that node invisible

7 Movement of the possessum is also possible if the possessum is a *wh*-item, e.g., *what*, without an NP-restriction, e.g., *Ti ehis dhiavasi tu Chomsky* 'lit. What have you read of Chomsky's?'.

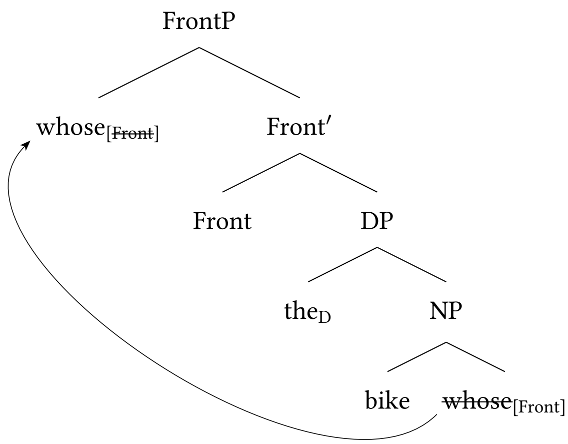
8 In minimalist terms, an intermediate projection corresponds to a syntactic object that is neither a maximal nor a minimal projection. Following Rizzi (2015), I adopt the Maximality Principle, according to which phrasal movement can target only maximal objects bearing a given label. It follows, therefore, that intermediate projections are ineligible for movement.

to the probe. In Keine's notation:

(12) Nominal Opacity (Greek): $[\text{wh/top/foc/rel}]_{C^0} \parallel N$

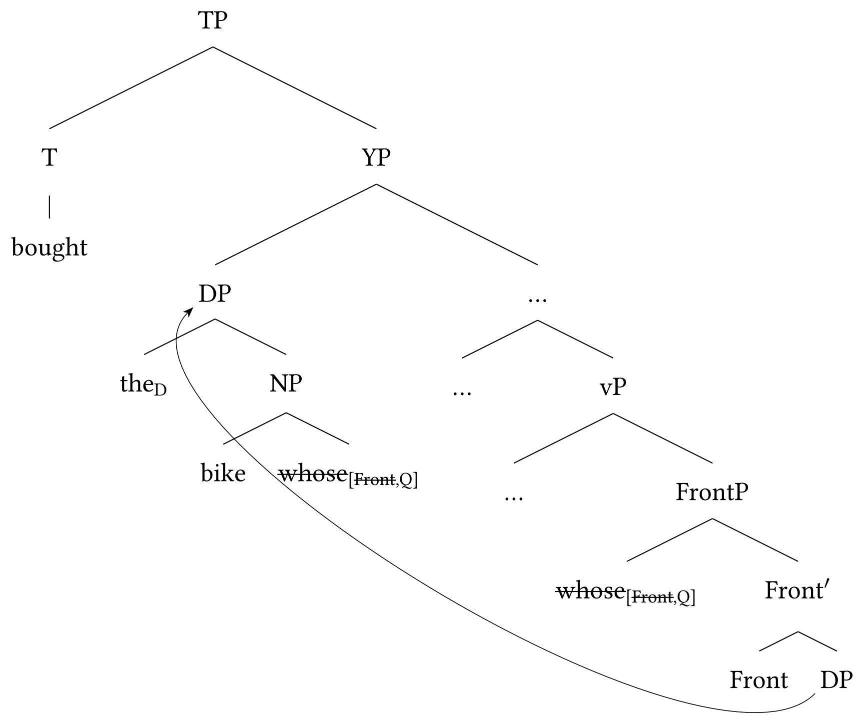
The only licit movement step of the possessor is that shown in (15), where it moves to Spec, FrontP. This is allowed because it constitutes movement within the extended projection of D, and is not motivated by C.

(13)



Subsequently, the DP remnant that contains the accusative possessum, *the bike*, is optionally allowed to undergo movement into a middle field position between the TP and the thematic position of external arguments, that is, the vP. This movement step of the possessum DP resembles regular accusative objects in Modern Greek. These objects can also move to this middle-field position, resulting in VOS word order. In this order, the object moves to a middle-field position—let us call it YP, as in (14)—lower than the verb in T but higher than the external argument which arguably stays in Spec, vP when it appears postverbally (Alexiadou and Anagnostopoulou 2001, Roussou and Tsimpli 2006, Alexiadou 2006, Anagnostopoulou 2003, Georgiafentis 2001, 2003 i.a.).

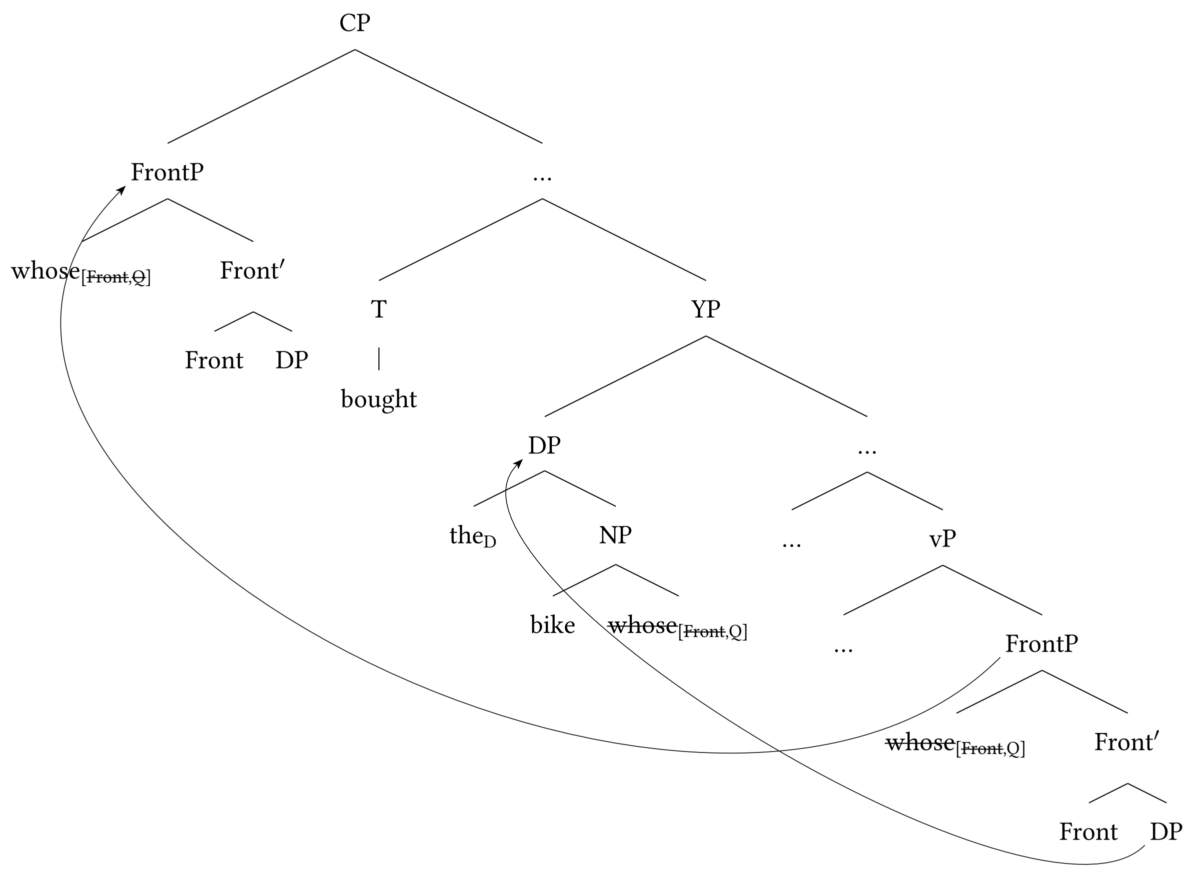
(14)



This movement step is compatible with our assumptions about nominal opacity in (12). Selective opacity is probe-specific, being defined with respect to a particular probe. In the present case, the relevant probe is the wh/relative/topic/focus feature on C^0. Opacity with respect to this probe does not entail opacity with respect to other features. Movement of the DP to the middle field is therefore expected to be possible under selective opacity, because, as will be discussed in the next section, it is an A-movement step triggered by a different probe, rather than by C bearing a *wh*-/focus/topic or relative feature.

The derivation below illustrates the three movement steps that underlie cases where the possessor occupies the left periphery. First, the possessor moves to Spec, FrontP. Next, the possessum DP moves to the middle-field, landing in Spec, YP, where objects are moved in VOS orders. Finally, the possessor undergoes further movement to the left periphery within the FrontP remnant, which gives rise to the surface order in (10a).

(15)



In this view, the derivations underlying (10a) and (10c) share a key similarity: in both cases, the entire FrontP moves to the left periphery of the clause. The crucial difference lies in whether the accusative possessum moves out of FrontP before this movement. In (10a), the possessum evacuates FrontP, resulting in the possessor being separated from the possessum. Conversely, if the possessum remains within FrontP, it moves together with the possessor to the left periphery. To explain these movements, I adopt the standard assumption that all syntactic movement is feature-driven (Chomsky 1993).[9] In (10c), the possessor carries two features: a 'Front' feature,[10] that triggers movement to Spec, FrontP, and a 'Q' feature, triggering movement to Spec, CP.[11] Similarly, the possessum must also carry a feature to license its movement to the middle-field. Notably, prenominal possessors in Greek obligatorily have a distinct emphatic interpretation (Ntelitheos 2004), often lacking in unfronted possessors (see 18a vs 18a for a fronted and an unfronted possessor). This suggests that movement to Spec, FrontP (i.e., the prenominal position) is motivated by interpretive reasons. Therefore, I analyze this movement as criterial movement (see discussion in Angelopoulos and Michelioudakis 2024 for evidence that Spec,FrontP is a criterial position), akin to movement to Spec, TopicP or Spec, FocusP. Consequently, in (15), the possessor, having undergone criterial movement to Spec, FrontP, is frozen in place, as a result of criterial freezing (see Rizzi 2006, 2010), and cannot move on its own. To satisfy its Q-feature, it must pied-pipe the entire FrontP to Spec,FocusP.

3.2 Middle-field movement

In this section, I present arguments for the view that plain possessor movement always involves the movement of the accusative possessum to the middle field,

[9] As an anonymous reviewer points out, there also exist alternative approaches where movement is not feature-driven (see Chomsky 2024).

[10] The feature's label is not important; if this left-peripheral position were FocusP, the feature could be reanalyzed as Focus.

[11] This Q-feature can alternatively be a Topic feature in cases where the element moved to the left periphery together with the preposition is not a *wh*-element but a plain DP.

and, crucially, that this movement constitutes an A-movement step. Moreover, the target position for these arguments is the same as that of objects in the VOS order. These assumptions are supported by new data from ditransitive constructions, particularly those containing a dative argument. In this applicative frame, the default order is dative preceding accusative (16a), though the reverse order is also possible (16b). A striking fact emerges in (16c) and (16d): possessor subextraction is systematically blocked from an accusative argument in ditransitive constructions, irrespective of whether the accusative appears before (16c) or after the dative argument (16d).

(16) a. Edhosan ton mathiton to vivlio tu Yalom.
gave.3PL the students.DAT the book.ACC the Yalom.GEN
'They gave students Yalom's book.'

b. Edhosan to vivlio tu Yalom ton mathiton.
gave.3PL the book.ACC the Yalom.GEN the students.DAT
'They gave students Yalom's book.'

c. * Pjanu edhosan ton mathiton [to vivlio t_i]?
whose.GEN gave.3PL the students.DAT the book.ACC
'Whose book did they give the students?'

d. * Pjanu edhosan [to vivlio t_i] ton mathiton?
whose.GEN gave.3PL the book.ACC the students.DAT
'Whose book did they give the students?'

If possessor splitting proceeded in a successive cyclic fashion through Spec,DP, as proposed by Horrocks and Stavrou (1987), both (16c) and (16d) should be grammatical, contrary to fact. In (16c), the possessor should be able to move through Spec,DP of the accusative argument and then past the dative argument, given that A-bar movement is generally possible across an A-position, much like A-bar movement across a subject. Likewise, in (16d), the possessor should be extractable via Spec,DP in the pre-dative position, unless no movement out of the accusative is possible in this derived position—an expectation consistent with the so-called freezing constraint.

The proposed analysis accounts for these facts more robustly: (16c) is ungrammatical because the accusative object remains in its base position, that is, it has not moved to the middle-field, as evidenced by its surface order after the dative. (16d) is also ungrammatical, despite the accusative object undergoing A-movement to the middle field, because A-movement across a dative object is independently known to be impossible (Anagnostopoulou 2003).[12] Crucially, Anagnostopoulou (2003) has shown that cliticization or Clitic Doubling of the dative argument obviates its intervention, thereby permitting A-movement of the accusative object past it. This suggests that cliticization of the dative should also facilitate possessor extraction from the accusative, as movement to the middle field should now be possible. Indeed, this prediction is borne out, (17).[13]

(17) ? Pjanu tus edhosan [to vivlio t_i]?
whose.GEN 3PL.M.DAT gave.3PL the book.ACC
'Whose book did they give them?'

Now a question is whether the possessum undergoes the same type of move-

12 A derivation in which the accusative object first undergoes A-bar movement past the dative and subsequently moves to the middle field via A-movement is also ruled out: the first step is A-bar movement (Anagnostopoulou 2003), making the second step—a transition from A-bar to A-movement—a clear case of improper movement, which is independently blocked.

13 For reasons that remain unclear, Clitic Doubling appears to degrade acceptability in (17) to some extent.

ment and to the same position as objects when they appear in the VOS order. There are different analyses for the VOS order in Greek. Some of them assume movement to the middle field (Alexiadou and Anagnostopoulou 2001, Roussou and Tsimpli 2006), whereas others take it that the object moves alongside the verb (Georgiafentis 2001). In support of the former, Alexiadou and Anagnostopoulou (2001) offers a compelling explanation for obligatory object movement in VOS constructions. They argue that in VOS orders, the subject occupies Spec,vP. Consequently, the object cannot remain in its base position because the VP cannot contain two DPs with unvalued case features at spell-out, requiring its movement. Since this movement step is due to case, it must also be an instance of A-movement, and, as we saw in (16), the accusative possessum also undergoes A-movement. Since both undergo the same type of movement step, it makes sense that objects in the VOS order as well as accusative possessums in plain possessor extraction move to the same position. Nevertheless, the motivation for this movement step seems to be different in the two cases. In particular, as we are going to see in the next section, the accusative possessum undergoes the same middle-field movement step in PP-splits as well. Nevertheless, the possessum in this case originates in a case position—the P-complement position—so its movement to the middle-field cannot be for case, in contrast to the movement of accusative objects in the plain VOS cases. This in turn suggests that middle-field A-movement in Greek is not uniformly motivated by case reasons. Instead, it seems plausible that Greek also has scrambling, which is similarly not always case-related. While the precise feature motivating scrambling remains an open question, with some works proposing a dedicated scrambling feature (Müller 1997, 1998 i.a.), I will assume that the same feature underlies the movement observed in the derivation underlying PP-splits.

3.3 A syntactic analysis of PP-splits

Let us consider now how we can extend this analysis to analyze the patterns found with PPs. We begin with a baseline sentence in (18a). As shown in (18b), the possessor can appear before the noun even in those cases where the noun is embedded under a P. However, (18c) illustrates that the possessor cannot occur on its own clause initially. Interestingly, these PPs also allow for PP-splitting, as demonstrated in (18d), in which case P and the possessor are moved to the left periphery, and appear separated from the possessum, which in turn appears post-verbally.

(18) a. Harike ja tin epithia tis Marias.
was.happy.3SG for the success.ACC the Maria.GEN
'She was happy for Maria's success.'

b. Harike ja tis Marias tin epitihia.
was.happy.3SG for the Maria.GEN the success.ACC
'She was happy for Maria's success.'

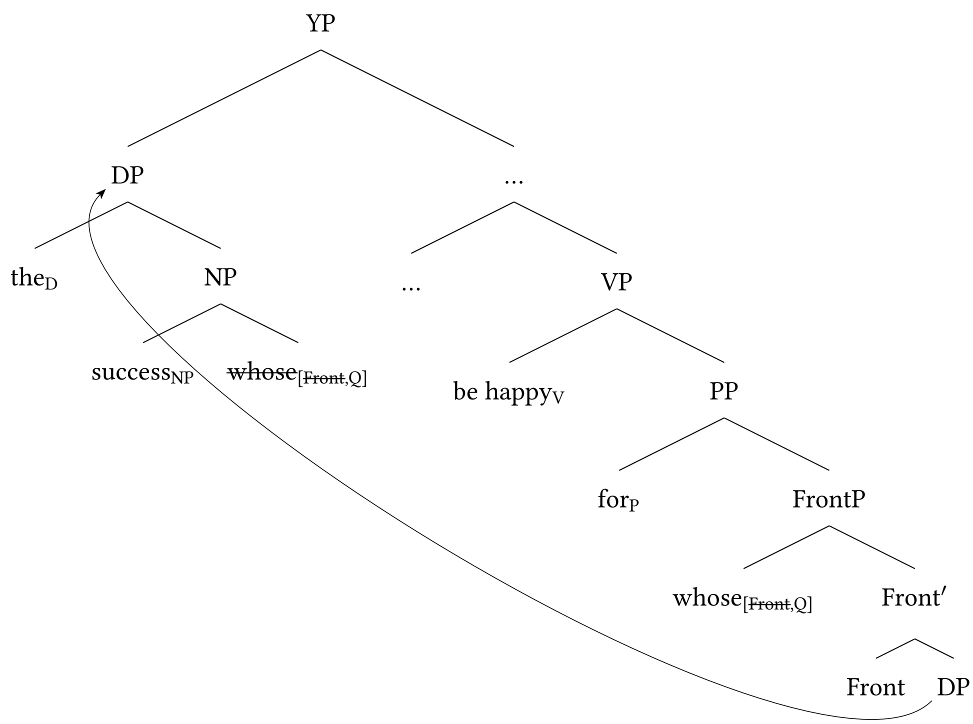
c. * Pjanu harike ja tin epitihia?
whose.GEN was.happy.3SG for the success.ACC
intended: 'For whose success is she happy?'

d. Ja pjanu harike tin epitihia?
for whose.GEN was.happy.3SG the success.ACC
intended: 'For whose success is she happy?'

Following Pesetsky (1996), I take PPs expressing subject matter like the ones in (18) to be externally merged in the V-complement position. FrontP is also available in this environment that allows for a prenominal possessor (18b). However, since the possessor undergoes criterial freezing in Spec,FrontP, it can never be extracted out of a PP, thus accounting for (18c) and the second row of Table 1, which illustrates the same fact for all PPs. On the other hand, after movement of the possessor to Spec,FrontP, where it satisfies its Front feature, the DP remnant comprising the possessum, *tin epitihia* 'the success,' is allowed to undergo movement into the middle-field. This movement step into the middle-field may involve an additional movement step through the PP specifier (not shown in 19). This is predicted if, following Abels (2003), Ps in languages that do not allow P-stranding like Greek are phase heads.[14]

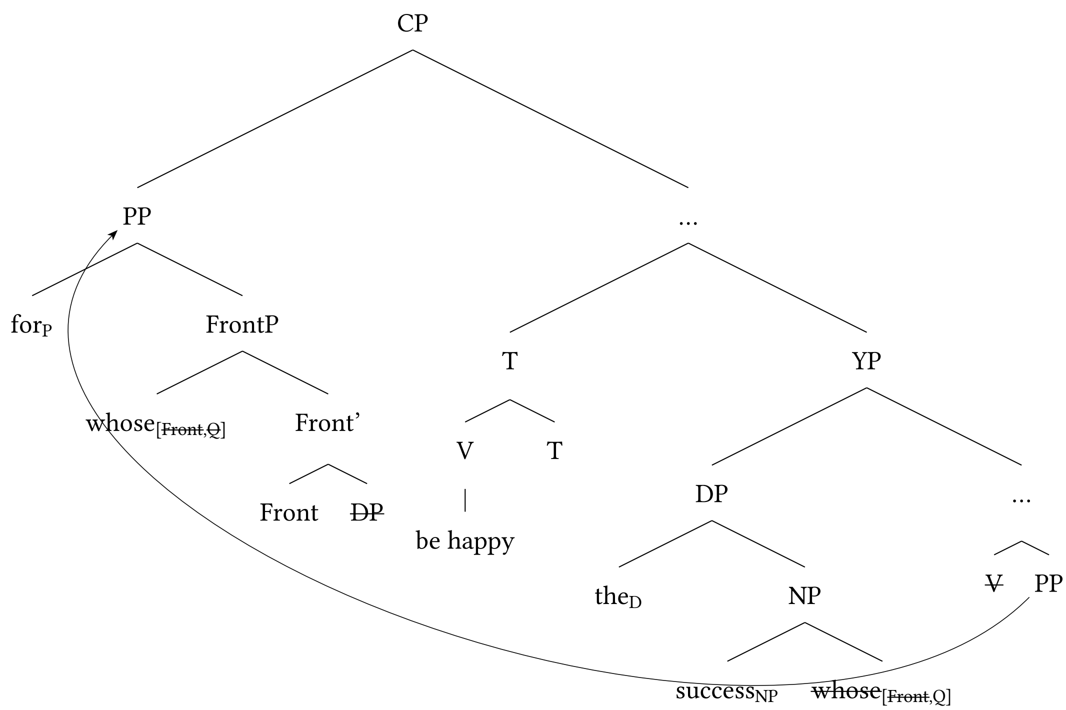
[14] Complex Ps, like *simfona me* 'according to,' realize a more complex structure. They possibly have a P-shell structure comprising P and little-p. Alternatively, *simfona* is a P, and *me* is the head of a case phrase, [$_{PP}$ P KP] (see Ramadanidis 2022 for a similar analysis of complex locative PPs). This would mean that *me* or *ja*, when they occur on their own, are K-heads but form a PP that includes a silent P.

(19)



The possessor also has a Q-feature that it needs to satisfy after moving to Spec,FrontP. Nevertheless, because the possessor has undergone criterial freezing in Spec,FrontP and therefore cannot move out of this position, its Q-feature can only be satisfied by pied-piping the PP remnant to the clause's left periphery, as illustrated in the structure below. Crucially, movement of FrontP alone, stranding P, is impossible because P is a phase in Greek, so this requires FrontP to move from the complement of P to its specifier, a step ruled out by Antilocality (Abels 2003). This correctly predicts why the P must move with the possessor to the left periphery.

(20)



To summarize, the analysis in (20) proposes that the ability of a PP to undergo PP-splitting is determined by its merge height relative to a middle-field position, YP in (19). Specifically, PP-splitting is permitted when a PP originates lower than YP, allowing A-movement of the accusative possessum to that position. This prediction is supported by the fact that benefactive, agent, causer and subject matter PPs are argumental PPs, thus they are E-merged within the vP, lower than YP. Manner PPs, similar to manner adverbials, are standardly assumed to be licensed in the vP/VP area (Cinque 1999, Alexiadou 1997 i.a.). Likewise, Michelioudakis and Angelopoulos (2019) have shown that instrument and comitative PPs are licensed by a verbal projection responsible for an eventive interpretation. Lastly, directional PPs are typically assumed to merge as arguments. In contrast, PPs merged above the middle field—high PPs—are predicted to resist splitting. Unlike low PPs (merged below YP), high PPs attach in the C-domain above TP, where the only possible probe is C. The accusative possessum thus cannot evacuate the DP where it originates, as movement out of the DP cannot be triggered by a C-probe (see nominal opacity in 12). PP-splitting is therefore blocked, since it requires a movement step of the accusative from inside the DP where it originates—a step impossible with high PPs. I propose that this is the case for evidential and temporal PPs, which I argue are merged above TP. This hypothesis aligns with existing research demonstrating that PPs occupy different syntactic positions depending on their interpretation (Alexiadou and Anagnostopoulou 2007, Cinque 2006, Schweikert 2005). In this literature, a standard assumption is that evidential and temporal PPs are merged higher than benefactive, source, agent, and other PPs allowing PP-splitting. (21) illustrates where the different PPs are merged relative to YP is provided below:

(21) Evidential, Temporal>**YP**>Benefactive, Source, Agent, Causer, Manner, Instrument, Locative, Comitative, Subject Matter

The two central components of the analysis, namely, that P-splits involve a multi-step derivation and that PPs are merged hierarchically, are supported from the behavior of different types of genitives and from Condition C effects. We begin with the genitives. The analysis predicts that only genitives that can undergo the initial DP-internal fronting to Spec,FrontP should be able to form a P-split. This prediction is borne out. Consider non-possessive, appositive genitives like *ton Vrixelon* in (22a). Unlike true possessors, these genitives cannot be moved to a pre-nominal position, as shown by the ungrammaticality of (22b) (Angelopoulos and Michelioudakis 2024).[15]

(22) a. Efije apo tin poli ton Vrixelon.
left.3SG from the city.ACC the Brussels.GEN
'He left from the city of Brussels.'

b. * Efije apo ton Vrixelon tin poli.
left.3SG from the Brussels.GEN the city.ACC
'He left from the city of Brussels.'

Because this initial DP-internal fronting step to Spec,FrontP is a necessary precondition for creating the remnant DP, the analysis correctly predicts that such constructions cannot form a P-split. The ungrammaticality of (23) confirms this.

(23) * Apo ton Vrixelon efije tin poli.
from the.GEN Brussels.GEN left.3SG the.ACC city.ACC
'From city of Brussels, he left.'

In short, the inability of appositive genitives to undergo DP-internal fronting directly correlates with their inability to participate in P-splits, strongly supporting the claim that the former is a prerequisite for the latter.

The idea that PPs are merged in distinct syntactic heights finds support in Condition C data, which I present in the contrast below between evidential and comitative PPs. This contrast is replicated between evidential and temporal PPs, and the rest of the PPs that allow PP-splitting. Specifically, both evidential and

15 Some speakers find (22b) acceptable.

comitative PPs can occur clause initially. However, whereas evidential PPs usually prefer this position, comitative PPs only do so when containing a focused XP, e.g., *secretary* in (24b). This contrast already suggests that evidential PPs are E-merged higher than comitative PPs. In (24a) and (24b), both PPs contain a proper name. It can be coindexed with the subject of the clause, shown as pro, when within an evidential PP, but not when in a comitative PP. This aligns with the assumption that evidential PPs are merged in the left periphery; specifically, evidential PPs are base generated in the left periphery, above where pro is interpreted, i.e. in Spec,TP (Angelopoulos and Sportiche 2021). Because of this, the proper name within the evidential PP and pro do not c-command each other, thus, the two can be freely coindexed, as shown in (24a). In contrast, comitative PPs originate below the middle-field, and thus, lower than Spec,TP. Their presence in the left periphery, as in (24b), results from movement from their base position. Hence, a lower PP copy containing the proper name exists under Spec,TP, where pro is interpreted. Due to Condition C, the proper name and pro cannot be coindexed in (24b) since pro c-commands the proper name in the lower PP copy.[16,17]

[16] Talić (2019) proposes an analysis of PP-splits in Bosnian/Croatian/Serbian, where the P cliticizes to a constituent that has moved to Spec, PP. Extending this to Greek would imply that the PP-split is created via movement of the possessor to Spec,PP, followed by cliticization of the P into the possessor. However, this account, while potentially suitable for Bosnian, faces a challenge when applied to Greek. As shown in (9b), Greek allows PP-splits involving complex Ps, which cannot be cliticized.

[17] There is a particular data point one must acknowledge that the proposed analysis cannot immediately account for. Let us consider the sentence below:

(1) * Harika tin epitihia ja tis Marias.
was.happy.1SG the success.ACC for the Maria.ACC

(24) a. Kata ti ghramatea tu $Jani_i$, $pro_{j/i}$ eprepe na ehi
per the secretary.ACC the John.GEN must na have.3SG
katethesi tin dhilosi pio prin.
submitted the form.ACC more early
'According to John's secretary, he must have submitted the form earlier.'

b. Me ti GHRAMATEA tu $Jani_i$, $pro_{j/*i}$ eprepe na ehi
with the secretary.ACC the John.GEN must na have.3SG
pai s-tin ekdhilosi.
gone to-the event
'With John's secretary, he must have gone to the event.'

4 *Distributed deletion

Several analyses assume distributed deletion to account for nominal splits like those examined here (Fanselow and Ćavar 2002, Pereltsvaig 2008, and Bondarenko and Davis 2023 i.a.). However, a challenge for distributed deletion accounts is determining the conditions under which it applies as well as why it applies the way it does.

Fanselow and Ćavar (2002) propose that splits arise only in the context of operator movement, specifically connecting split NPs to information structure.

'I was happy for the success of Maria.'

Under the proposed analysis, this should be a possible structure; it corresponds to a derivation where the PP is in-situ but the accusative possessum has moved out of it to the middle-field. This does not immediately speak against the proposed analysis. One should consider the interpretive effects that arise in this configuration; it is possible that movement of the possessum to the middle-field brings about a particular interpretation of the DP inside the PP that can only be encoded in the left-periphery thus requiring its movement to the left periphery.

This approach proposes that if a constituent XP, composed of a head X and e.g., an NP-complement, carries distinct features (e.g., [Topic] on X and [Focus] on NP), then the entire XP moves as a single unit to distinct feature-checking positions (e.g., Spec,TopP and Spec,FocP), as illustrated in (25). Subsequently, X and NP are spelled out in the position where they have their feature checked. Consequently, in (25), the NP is spelled out in the lower XP copy, while X is spelled out in the high XP copy, yielding a split.[18]

(25) $[[_{\text{FocP}}\ [_{\text{XP}}\ \mathbf{X}_{[\text{Foc}]}\ \text{NP}_{[\text{Top}]}\ ]\ \text{Foc}_{[u\text{Foc}]}\ \ldots\ [_{\text{TopP}}\ [_{\text{XP}}\ \text{X}_{[\text{Foc}]}\ \mathbf{NP}_{[\text{Top}]}\ ]\ \text{Top}_{[u\text{Top}]}\ \ldots]$

While applying (25) to derive an NP-split seems straightforward, deriving a PP-split requires additional principles. For example, Fanselow and Ćavar (2002) examines the Croatian split in (26a). Deriving this split requires two movements of the PP, *na kakvo stablo* 'what kind of tree,' from its base-generated position, as illustrated in (26b). *Stablo* 'tree' checks a feature in the intermediate position, while *nkakvo* 'kind of' checks a feature in the left periphery. *Stablo* is spelled out in the intermediate copy where it satisfies its feature, and *kakvo* is spelled out in the left periphery where it satisfies its own feature. To account for the obligatory adjacency of the P with *kakvo*, an additional principle is posited: a P must be phonologically adjacent to its selected category (see Goncharov 2015 for criticism and a syntactic alternative for PP-splits in Slavic languages).

[18] Hinterhölzl (2000, 2002) argues that only pied-piped material, and not the movement-triggering feature itself, can be deleted in the higher copy (see also van Urk 2024 for a similar proposal).

(26) a. Na kakvo se Ivan stablo penje?
on what-kind-of self I. tree climbs
'On what kind of tree does Ivan climb?' Fanselow and Ćavar (2002, (96))

b. [$_{PP}$ Na kakvo ~~stablo~~] se Ivan [$_{PP}$ ~~na kakvo~~ stablo] penje [$_{PP}$ ~~na kakvo stablo~~].

The conditions for licensing a split proposed by Fanselow and Ćavar (2002) are also met in the Greek PP-splits we analyze. Specifically, similar to the XP structure in (25), the PP comprises two parts, the possessum and the possessee, each carrying a formal feature. Therefore, we would expect PP-splits under this view to be derived from distributed deletion. To illustrate how this would work, consider the PP-split in (18d), repeated below:

(27) Ja pjanu harike tin epitihia?
for whose.GEN was.happy.3SG the success.ACC
intended: 'For whose success is she happy?'

Under a distributed deletion approach, three copies of the PP would be generated: one in its base position, the V-complement position, one in the middle field, where the accusative possessum checks a feature also involved in the movement of object in VOS, and a third in the left periphery, where the possessor checks its Q-feature. Distributed deletion would then apply, spelling out the possessor and possessum in the positions where they check their features, resulting in the split.

(28) [$_{CP}$ [$_{PP}$ for ~~the success~~ whose] [$_{\dots}$ [$_{TP}$ be happy$_{T}$ [$_{PP}$ ~~for~~ the success

~~whose~~] [... [$_{VP}$ V [$_{PP}$~~for the success whose~~]]]]]

A question that arises in light of (28) is why the P is spelled out high together with the possessor. Our proposed account offers a straightforward explanation for this phenomenon (see discussion preceding 20). However, under Fanselow and Ćavar's (2002) approach, we would expect the P to be spelled out next to its selected complement, *the success*, in the intermediate copy. Therefore, this observation presents initial evidence that a distributed deletion analysis yields incorrect predictions. Now, consider the ungrammatical sentence in (18c), repeated below for convenience:

(29) * Pjanu harike ja tin epitihia?
whose.GEN was.happy.3SG for the success.ACC
intended: 'For whose success is she happy?'

The ungrammaticality of this example follows from the proposed account because possessors undergoing criterial freezing inside the PP, cannot be extracted out of it (see discussion preceding 20). In contrast, a distributed deletion account incorrectly predicts it to be grammatical. Under such an account, the possessor, *whose*, would be spelled out in the left periphery, where its Q-feature is checked, as shown in (30). The accusative possessum, *success*, would be spelled out in the middle field, checking the feature associated with object movement in VOS orders. The P would also be pronounced in the middle field, adhering to Fanselow and Ćavar's principle that Ps must be spelled out next to the element it selects.

(30) [$_{CP}$ [$_{PP}$ ~~for the success~~ whose] [$_{...}$ [$_{TP}$ be happy$_{T}$ [$_{PP}$ for the success ~~whose~~] [... [$_{VP}$ V [$_{PP}$ ~~for the success~~ whose]]]]]

We next turn to the limitations of a more recent alternative proposed by Bondarenko and Davis (2023), who formalize distributed deletion as follows:

(31) PF rule for distributed deletion

a. When a moved NP or PP contains an element bearing a feature *F* that encodes contrastive focus or topicalization, realize only the element bearing *F* in the highest copy of the movement chain, and realize all other nodes in the lowest copy of the movement chain.

b. $[_{NP}\ XP_F\ \cancel{YP}\ \cancel{N_1}]$ … $[_{NP}\ \cancel{XP_F}\ YP\ N_1]$ Bondarenko and Davis (2023, (53))

This PF rule faces several theoretical problems. First, it remains unclear why it applies only to NPs and PPs, what prevents it from targeting CPs or VPs as well? How can a PF rule see syntactic categories and apply to a subset of them? The proposal offers no explanation. Second, the semantics it presupposes are problematic: why should the rule be limited to contrastive focus and topicalization, but not extend to plain focus?

Even setting these theoretical issues aside, the rule makes incorrect empirical predictions. In the case of PPs, since Ps do not bear features like Topic or Focus, the same problem arises as in the earlier account: what determines the realization of the preposition? Furthermore, consider appositive genitives, which, as previously discussed, cannot be fronted prenominally (i.e., to Spec,FrontP), as shown in (32a) and (32b) (repeated from 22a and 22b). These genitives also fail to form P-splits, as shown in (32c) (repeated from 23). This presents an important problem for the PF rule above, as it should be able to apply and thus derive a P-split, just as with other PPs.

(32) a. Efije apo tin poli ton Vrixelon.
left.3SG from the city.ACC the Brussels.GEN
'He left from the city of Brussels.'

b. * Efije apo ton Vrixelon tin poli.
left.3SG from the Brussels.GEN the city.ACC
He left from the city of Brussels.'

c. * Apo ton Vrixelon efije tin poli.
from the Brussels.GEN left.3SG the city.ACC
'From the city of Brussels, he left.'

The PP in (32c) contains a DP with a contrastive-focus interpretation, *ton Vrixelon*. If P-splits were derived via the same PF rule, we would expect this PP to form a P-split as well, just like other PPs (see Table 1): the P and the contrastive focused DP should be able to be spelled out in the left periphery stranding the possessum postverbally, However, this prediction is incorrect. Consequently, the main shortcoming of the PF-based account is that it fails to capture the observed correlation between the availability of prenominal genitives and the possibility of P-splitting—a correlation that the present proposal successfully explains.

Other accounts of distributed deletion face similar issues: they fail to generalize beyond the languages they were designed for, or rely on additional stipulative mechanisms. This is, for instance, the case for Iquito, where a distributed deletion analysis has been proposed (Murphy and Wilson 2025). However, this account is highly language-specific and cannot be extended to Greek or other languages. Moreover, it requires an additional diacritic feature to operate. We will not discuss this proposal further, as it has limited relevance for cross-linguistic comparison.

An anonymous reviewer notes that the lack of cross-linguistic generalizability of distributed deletion is not necessarily problematic, since languages are expected to differ in how their syntactic structures are externalized, which might allow distributed deletion to operate differently across languages.[19] This is in principle correct. Under a single mechanism of copy deletion, for instance, cross-linguistic differences in PF conditions may determine whether the higher copy, the lower copy, or both copies are spelled out. A similar scenario could in also hold for distributed deletion if the same operation (e.g., the version proposed by Fanselow and Ćavar 2002) applied universally, with languages differing only for example, in whether the relevant feature that creates the condition for distributed deletion is Topic in one language and Focus in another.

However, this is not the situation we find with distributed deletion. In existing proposals, split constituents in different languages are analyzed as the result of entirely different distributed deletion mechanisms, none of which can be straightforwardly extended to other languages. As discussed above, some accounts assume that the XP carrying the movement-triggering feature must be spelled out, while others assume the opposite, that all XPs except the one bear-

[19] Likewise, an anonymous reviewer suggests that a potential weakness of the paper is that it does not demonstrate how the proposed account could be extended to split constructions in other languages. This, however, is not the goal of the paper. The aim is not to show that the specific analysis proposed here, based on remnant movement, accounts for all instances of splitting cross-linguistically. Rather, the central claim is that split constructions must be derived by purely syntactic means and, crucially, that distributed deletion should be excluded as a possible mechanism for split formation.

ing the movement-triggering feature are pronounced, and still others introduce additional machinery, such as diacritic features. Some of them tie the application of distributed deletion to information structure properties whereas others do not. The limitation of distributed deletion, therefore, is that no single mechanism of it can be shown to underlie split constituents cross-linguistically.

One could alternatively assume that languages do not differ in how distributed deletion operates, but rather in whether they allow the operation at all. On this view, Greek may simply lack distributed deletion, while other languages have it. However, this scenario is not compatible with how distributed deletion has been proposed in the literature. Under current proposals—especially those linking the operation to Topic or Focus—the prediction is that distributed deletion should in principle be available in all languages, since such information-structural properties appear to be universally available. Moreover, nothing in the proposed conditions on distributed deletion suggests a principled way in which the operation could be active in one language but entirely absent in another. Consequently, if we examine a language that satisfies the conditions under which distributed deletion is predicted to apply and show that its split constructions cannot be derived through this operation—as argued here for Greek—this constitutes strong evidence against distributed deletion as a grammatical mechanism more generally.

To sum up, the proposed analysis fares better than a distributed deletion alternative because it only works with what is independently needed; a general PF-mechanism of copy deletion that applies to every instance of internal merge.[20] No additional PF-mechanisms that only derive splits are needed, nor are additional assumptions required to account for where, for example, a P is spelled out, nor additional spurious features, such as a strong P-feature that is important in Fanselow and Ćavar (2002) to motivate movement of the P. Furthermore, the principles that apply to distributed deletion lack theoretical grounding or are very specific to certain languages or constructions, which in turn make a cross-linguistic analysis not possible. For instance, Fanselow and Ćavar's PF principle governing P spell-out faces this challenge: it is not clear why a PF principle should be sensitive to P's selectional requirements. Also, as we saw, this principle does not hold in Greek. Furthermore, as discussed in Murphy and Wilson (2025), Fanselow and Ćavar's distributed deletion (or other versions of distributed deletion) account cannot be extended to Iquito splits. This has as a consequence that a new, language-specific distributed deletion mechanism with novel constraints is needed. This approach raises a significant concern: the absence of a principled account of distributed deletion allows arbitrary modifications in its application and the postulation of novel PF-principles, thereby rendering the theory effectively unfalsifiable. In contrast, the proposed analysis does not face this challenge, as it assumes merge properties and syntactic hierarchies that should hold across languages, and should thus be able to be tested. The above data are strong evidence against distributed deletion in the formation of splits. Combined with

[20] See Collins and Stabler (2016) for a spell-out mechanism that can also handle remnant movement.

similar conclusions in Goncharov (2015), who proposes a syntactic alternative to the splits initially posited by Fanselow and Ćavar (2002) as evidence for distributed deletion, these concerns suggest that distributed deletion is not attested in natural language.

Before we close, a question to address if our claims so far are on the right track is why exactly distributed deletion is not allowed in natural languages. It is reasonable to assume that constituency plays an important role here. Syntax respects constituency and the picture that arises from deletion phenomena, such as ellipsis, is that constituency is respected there too, as only constituents can be deleted (Fox 2000, Merchant 2001 among others). Consequently, distributed deletion seems to be ruled out in natural languages simply because it is a mechanism that disregards constituency, permitting the deletion of non-constituents.

5 Conclusion

This paper investigated a split of Greek in which a possessor within a PP is separated, along with P, from the possessum. New data revealed that this split is sensitive to the PP's semantic interpretation. PPs are merged at distinct syntactic heights in accordance with their interpretation. Thus, the availability of PP-splitting is linked to the PP's merge height relative to an independently motivated middle-field position. However, a distributed deletion account proves inadequate. It not only requires supplementary principles to explain PP-splits but also yields incorrect predictions. Similar criticisms have been leveled against distributed deletion accounts of splitting phenomena in other languages (Goncharov 2015), thus refuting the existence of distributed deletion.

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